P. R. MADHEVAN

Firmware & SoC Systems Engineer — WLAN Silicon, Low-Power Architecture, Pre-Silicon Bring-up

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SUMMARY

Firmware engineer with six years on 802.11 WLAN silicon, working where firmware meets hardware: low-power SoC architecture, pre-silicon bring-up, and host-firmware transports over PCIe, SDIO and on-die AXI. Equally at home tracing a defect through RTL and waveforms, writing the Cortex-R4 / M4 / M52 firmware that works around it, and building the debug and automation infrastructure a bring-up runs on.

TECHNICAL SKILLS

Languages C, ARM assembly (Cortex-R4 / M4 / M52), Python, TCL, Bash

Firmware & OS FreeRTOS, bare-metal bring-up, embedded Linux kernel and device drivers, CMSIS, linker scripts and memory layout, secure boot, Yocto

Wireless 802.11 a/b/g/n/ac/ax (Wi-Fi 6 / 6E), power save (DTIM, PS-Poll, U-APSD), WPA2 / WPA3, CCMP and TKIP, Wi-Fi NAN, FTM / RTT ranging, DOCSIS, Hotspot 2.0

SoC & interconnect AXI / AHB / APB, PCIe, SDIO, I2C, SPI, UART, GPIO and pinmux, M2M DMA, PMU and clock / power domains, GCI, OTP, msgbuf ring protocols

Debug & validation JTAG and ARM ADIv5 (DP / AP, MEM-AP), OpenOCD, GDB, Lauterbach TRACE32, Veloce hardware emulation, FPGA, RTL and waveform analysis, Wireshark

Automation Jenkins CI, Git, Bitbucket, Makefiles, custom static and dynamic memory-analysis tooling

EXPERIENCE

embedUR Systems — Chennai, India **May 2020 – Present**

Technical Lead (2025–present) · Senior Software Engineer (2023–2025) · Software Engineer (2020–2023)

Windows WLAN Driver — client: Infineon (Midiana) *Jul 2026 – Present*

- Building the wl utility and DHD applications for the **Windows WLAN driver** on the Midiana platform.
- Evaluating distributed real-time location over **Wi-Fi NAN ranging and FTM / RTT** — DS-TWR against SS-TWR, leading-edge detection and EKF innovation gating.

WLAN Firmware & SoC Bring-up — client: Synaptics *Jun 2022 – Jul 2026*

- Designed the MCU-WLAN sleep protocol for the 5312 SoC so either core can sleep independently with **zero packet loss** across full-retention, memory-retention and standby modes — a four-state machine in which every transition moves exactly one core, with receive buffers retargeted from host memory to local TCM so the MAC keeps receiving while the MCU sleeps. Design complete and ~45% implemented when the engagement ended. Cortex-R4 and Cortex-M52 firmware over the on-die AXI link.
- **Filed an invention disclosure** covering retrieval of AP-buffered packets under high host-wake latency, by designing an adaptive U-APSD / PM-2 handler that fetches a bounded number of buffered frames into limited dongle memory while the host wakes, preserving firmware offload. 802.11 power save.
- Eliminated spurious interrupts across the MCU-WLAN link by root-causing them to a two-flop synchroniser deassertion re-cross between the **96 MHz and 192 MHz** clock domains, then deferring the clear-pending step so level-triggered semantics separate genuine edges from spurious ones. GCI, CMSIS, NVIC.
- Kept the M2M DMA engine usable despite a silicon AXI defect, by proposing a workaround algorithm that classifies 4 KB boundary crossovers and 3-6 byte fragments at runtime so only the unsafe transfers reroute through memcpy; implemented with another engineer.
- Proved out the 5312 host-firmware data path on a **single DMA channel**, by building the proof of concept for re-hosting the PCIe msgbuf ring protocol over the internal AXI link as SCI; the team carried it to production.
- Enabled roughly **90% of 5312 development to proceed before first silicon**, by repurposing the on-die Bluetooth Cortex-M4 as the MCU host on existing dongle silicon and writing startarmcm4.S, the FreeRTOS port and the SCI driver from scratch; proposed the configuration as a new reduced-footprint MCU + Wi-Fi product segment.
- Gave the 5312 program hardware debug access on emulation by bringing up **JTAG and OpenOCD on the Veloce emulator**, root-causing four RTL defects (TCK skew from PnR routing, DBGPWRUPREQ not powering the PMU, ARM debug module held in reset, TCK-bound fetch speed) and proposing the fixes, including a new PMU control register. ARM ADIv5, RTL and waveform analysis.
- Let **existing register-level validation scripts run unchanged** on an SoC with no JTAG, SDIO, PCIe or USB transport, by extending the customer's verification environment to reach the chip over OpenOCD; adopted by the validation teams and later reused on FPGA.
- Drove 5312 Veloce bring-up **from reset to ping**, and contributed the architectural analysis behind a **2x SCI throughput** improvement and a shorter MCU boot.

- Cut active power **~70%** by proposing and implementing DTIM-based power reduction, and shipped SVT releases for 43711a0, 4362a2 and 4612a0 on time for an Amazon demo by porting the sdiopoolbank RAM-bank turn-off feature — a 960 KB target reached with a three-pass linker relocation of packet pools.
- Restored throughput on stalled SDIO links by diagnosing a bidirectional-traffic stall as an RX pool-starvation loop from its counter signature (idle pool 0, SD Rxfill 0, host RX backlog 181), then reserving a floor of packet pools exclusively for RX DMA refill so the release side always progresses.
- Closed the pairwise TKIP attack surface without breaking mixed legacy networks, by blocking TKIP on the pairwise path across the supplicant, DHD and firmware MAC key-install code while permitting it as the group cipher. WPA2, CCMP.
- Sped the Veloce emulator **~400%**, compressing a ROM tapeout cycle to **one week** and lifting emulator utilisation to near 100%, by rebuilding WLAN automation as Vista and extending it into a nightly regression suite across bands, channels and security modes. Python, TCL, Jenkins CI.
- Wrote the debug tooling used through 5312 pre-tapeout: **Marbel** (reconstructs system state from raw memory dumps via a Rabin-Karp scan and a bl/blx back-check stack unwinder), **Hammer** (dynamic heap flame charts), **Spade** (static struct padding and cache-line analysis), **Crisp** (firmware preload) and **Spark** (CH347 strap-pin automation with high-Z contention avoidance, enabling fully remote CI reflash).
- Found **200+ potential crash points** by reverse-engineering disassembled firmware for illegal UDEF instructions and proposed the fix; separately restored system discovery by correcting an MCU EROM parser inconsistency exposed once a second core was added.
- Resolved a **day-0 customer escalation** (BTSDIO / Ampak) by programming internal GCI interrupt generators and the on-chip JTAG master to reach UDR registers.
- Mentored **two engineers** through bring-up and low-level debug work; both went on to earn customer appreciation for their contributions.

Embedded Linux & Industrial Switching — client: Hirschmann (Belden) *May 2021 – Jun 2022*

- Cut boot time on GRS and MSP switches **from 165-172 s to 40-45 s** during the VxWorks-to-Linux migration, by tracing the delay to a 10-second retry loop inside hwscan_ethernet_module_info_check() and moving module initialisation into two dedicated tasks communicating over IPC, mirroring the VxWorks design. **Pat-on-the-back award from the customer and manager.**
- Fixed module-load failures at boot by establishing that the kernel BDE was not modprobed ahead of the user BDE — **the customer cited the depth of the analysis**. Linux kernel, device drivers.
- Sped module loading with a new algorithm supporting Hirschmann's pluggable module cards. **Spot award.**
- Contributed to hot-pluggable SFP and PSU module support on the Dragon, GRS1040 and MSP40 platforms — a Linux I2C EEPROM driver written on the Hirschmann I2C adapter to read 27 module parameters through a two-stage CPLD multiplexer, plus device-tree changes, a temperature-sensor race fix and a PCIe enumeration-order correction — supporting the engineer leading that work.

Cable Modem Firmware & Lab Provisioning — client: CommScope (ARRIS) *May 2020 – May 2021*

- Debugged DOCSIS signal-strength, DHCP, Green Ethernet and connection defects on SURFboard G34 gateways, across the SNMP and dmcli backends. RDKB, Yocto.
- Cut client test-machine cost **~90%** during a hardware shortage by building a Raspberry Pi replacement for the provisioning client, and brought up the ARRIS lab provisioning system and CMTS server behind it.

EDUCATION

B.E. Electrical & Electronics Engineering — SRM Easwari Engineering College, Anna University, Chennai *2016 – 2020*

- **Gold Medalist, summa cum laude.**
- Higher Secondary **94%** · Secondary **93%** — Sri Sankara Vidyashramam Matriculation Higher Secondary School, Chennai.
- Final-year project: **mobile communication over TV White Space (TVWS)** — drove transmit and receive directly from Raspberry Pi GPIO pins instead of dedicated RF transceiver chips, with Reed-Solomon forward error correction and POCSAG signalling.

RECOGNITION

- **Invention disclosure filed** — controlled AP-buffered packet retrieval with adaptive U-APSD control (Synaptics).
- **Customer award** for the Veloce emulator optimisation (Synaptics) · **pat-on-the-back award** from the customer and manager for the switch boot-time reduction (Hirschmann) · **spot award** for the pluggable-module algorithm (Hirschmann) · multiple Synaptics Hi-Fives.
- Rated **5 / 5, "significantly exceeded job role"**, for three consecutive annual review cycles.